

# Advanced Lesson 2 - Exercises: Exact and Approximate Arithmetic

## Mikey Ferguson

### MATH 7013 - Advanced Mathematical Modeling

```
Clear[a, b, c, d, i, n, f, g, h];
```

---

1: How many digits of accuracy does each expression carry?

$$\left\{ \frac{3}{7}, \text{Precision}\left[\frac{3}{7}\right] \right\}$$

$$\left\{ 2 + \frac{3}{7}, \text{Precision}\left[2 + \frac{3}{7}\right] \right\}$$

$$\left\{ \text{InputForm}\left[2.\text{' + } \frac{3.\text{'}}{7.\text{'}}\right], \text{Precision}\left[2.\text{' + } \frac{3.\text{'}}{7.\text{'}}\right] \right\}$$

$$\left\{ \text{InputForm}\left[1.23\text{' + } \frac{1.2345\text{'}}{1.234567\text{'}}\right], \text{Precision}\left[1.23\text{' + } \frac{1.2345\text{'}}{1.234567\text{'}}\right] \right\}$$

$$\left\{ \text{InputForm}\left[1.2\text{' + } 1.234567890123456789\text{' 19.0915149772127 + } 1.2345678901234567890123456789\text{' 29.091514977212704}\right], \right. \\ \left. \text{Precision}\left[1.2\text{' + } 1.234567890123456789\text{' 19.0915149772127 + } 1.2345678901234567890123456789\text{' 29.091514977212704}\right] \right\}$$

$$\left\{ \text{InputForm}\left[1.234567890123456789\text{' 19.0915149772127 + } 1.2345678901234567890123456789\text{' 29.091514977212704}\right], \right. \\ \left. \text{Precision}\left[1.234567890123456789\text{' 19.0915149772127 + } 1.2345678901234567890123456789\text{' 29.091514977212704}\right] \right\}$$

$$\left\{ \text{InputForm}\left[N\left[\frac{2}{7}\right]\right], \text{Precision}\left[N\left[\frac{2}{7}\right]\right] \right\}$$

$$\left\{ N\left[\frac{2}{7}, 100\right], \text{Precision}\left[N\left[\frac{2}{7}, 100\right]\right] \right\}$$

$$\left\{ \sqrt{2}, N\left[\sqrt{2}\right] \right\}$$


---

## 2: Six ways to sum $1/n$ from 1 to 100

$$\sum_{n=1}^{100} \frac{1}{n}$$

$$\sum_{n=1}^{100} \frac{1.}{n}$$

$$\sum_{n=1}^{100} N\left[\frac{1}{n}\right]$$

$$N\left[\sum_{n=1}^{100} \frac{1}{n}\right]$$

$$N\left[\sum_{n=1}^{100} \frac{1}{n}, 50\right]$$

$$N\left[\sum_{n=1}^{100} \frac{1.}{n}, 50\right]$$


---

## 3: $E^{\pi}$ to 500 digits, and exactly

$$N[e^{\pi}, 500]$$

$$e^{\pi}$$


---

## 4: A complex sum

$$\sum_{n=1}^{10} \frac{1}{n + 2 \, i}$$

$$N\left[\sum_{n=1}^{10} \frac{1}{n + 2 \, i}, 50\right]$$

$$N\left[\sum_{n=1}^{10} \frac{1}{n + 2. \, i}, 50\right]$$


---

## 5: Sin[6] exactly and approximately

$$\text{Sin}[6]$$

```
Sin[6.`]
N[Sin[6], 50]
N[Sin[6.`], 50]
N[Sin[N[6, 20]], 50]
```

---

## 6: The partial sums as a user function

```
Clear[f, n, nTerms];
f[nTerms_] :=  $\sum_{n=1}^{nTerms} \frac{1}{n}$ 
{f[1], f[10], f[100]}
```

---

## 7: How many system functions are there?

Names["System`\*"] returns the whole list of built-in names. It is far too long to print in a submitted notebook, so only its length is evaluated below; the list itself is computed and discarded.

```
Length[Names["System`*"]]
```

---

## 8: myDiff1 and myDiff2 - why differentiation needs delayed assignment

```
Clear[x, f];
myDiff1[f_] =  $\partial_x f$ 
{myDiff1[x2 + 3 x + 4], myDiff1[e2 x + 3], myDiff1[f[x]]}
?myDiff1

Clear[x, f];
myDiff2[f_] :=  $\partial_x f$ 
{myDiff2[x2 + 3 x + 4], myDiff2[e2 x + 3], myDiff2[f[x]]}
?myDiff2
```